

Name: _____

These questions assess your understanding of the material from Chapter(s) 25, 26 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. A clear-glass light bulb is placed 0.49 m from a thin convex lens that has a focal length of 0.051 m. Calculate the magnification of the produced image.

ANSWER: -0.116 ✓

$$d_o = 0.49 \text{ m}$$

$$f = 0.051 \text{ m}$$

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\frac{1}{d_i} = \left(\frac{1}{0.051 \text{ m}} \right) - \left(\frac{1}{0.49} \right) = \frac{1}{17.567} = 0.057 = d_i$$

$$m = \frac{-d_i}{d_o} = \frac{-0.057 \text{ m}}{0.49 \text{ m}} = -0.116$$

2. You shine a laser through air at an incident angle of 39° into an unknown material and measure the angle of refraction to be 33° . What is the unknown material's index of refraction?

ANSWER: 1.16 ✓

$$\theta_1 = 39^\circ$$

$$n_1 = 1$$

$$\theta_2 = 33^\circ$$

$$n_2 = ?$$

$$n_1 \frac{\sin \theta_1}{\sin \theta_2} = n_2 \frac{\sin \theta_2}{\sin \theta_2}$$

$$n_2 = \frac{1 \sin 39^\circ}{\sin 33^\circ} = 1.16$$

3. What power of spectacle lens is needed to correct the vision of a near-sighted person whose far point is 120 cm? Assume the spectacle lens is held 1.3 cm away from the eye by eyeglass frames.

ANSWER: -0.842 D ✓

$$d_o = \infty$$

$$d_i = 1.20 \text{ m} - 0.013 \text{ m}$$

$$d_i = -1.187 \text{ m}$$

$$P = \frac{1}{d_o} + \frac{1}{d_i} = -0.842 \text{ D}$$

4. What is the size of the image on the retina of a 0.60 mm-diameter strand of hair that is held at arm's length (72. cm) away? Take the lens-to-retina distance to be 1.7 cm.

ANSWER: $-1.42 \times 10^{-5} \text{ m}$

$$h_o = 0.60 \times 10^{-3} \text{ m}$$

$$d_o = 0.72 \text{ m}$$

$$d_i = 0.017 \text{ m}$$

$$h_i = ?$$

$$m_o \cdot \frac{h_i}{h_o} = -\frac{d_i}{d_o} \cdot h_o$$

$$h_i = -\frac{0.017 \text{ m}}{0.72 \text{ m}} (0.60 \times 10^{-3} \text{ m})$$

$$h_i = -1.42 \times 10^{-5}$$

$$n_1 > n_2$$

5. Calculate the critical angle for light traveling in a material that is surrounded by air, given that the material has an index of refraction of 1.7.

ANSWER: 36.03°

$$n_1 = 1.7$$

$$n_2 = 1$$

$$\frac{n_1 \sin \theta_c}{n_1} = \frac{n_2}{n_1}$$

$$\sin \theta_c = \frac{1.00}{1.7}$$

$$\sin \theta_c = 0.588$$

$$\theta_c = 36.03^\circ$$

6. A scientist at a natural-history museum is holding a specimen 6.3 cm from a thin convex magnifying glass of focal length 18. cm. What magnification is produced?

ANSWER: 1.54

$$d_o = 0.063 \text{ m}$$

$$f = 0.18 \text{ m}$$

$$m = ?$$

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\frac{1}{d_i} = \left(\frac{1}{0.18} \right) - \left(\frac{1}{0.063} \right) = -\frac{1}{0.097}$$

$$d_i = -0.097 \text{ m}$$

$$m = -\frac{d_i}{d_o} = \frac{0.097 \text{ m}}{0.063} = 1.54$$

43

7. Calculate the speed at which light travels in a particular material given that the index of refraction of the material is 1.3.

ANSWER: $2.308 \times 10^8 \text{ m/s}$

$$c = 3 \times 10^8 \text{ m/s}$$

$$n = 1.3$$

$$v = ?$$

$$\frac{v}{c} = \frac{1}{n}$$

$$v = \frac{c}{n} = \frac{3 \times 10^8 \text{ m/s}}{1.3} = 2.308 \times 10^8$$

8. Suppose that you take a magnifying glass out on a sunny day and you find that it concentrates sunlight to a small spot 5.8 cm away from the lens. What is the power of the lens?

ANSWER: 17.2 D

$$d_o = 0.058 \text{ m}$$

$$P = \frac{1}{d_o} = \frac{1}{0.058} = 17.2 \text{ D}$$

9. Calculate the power of the eye when viewing an object 2.7 km away. Take the lens-to-retina distance to be 2.5 cm.

ANSWER: 40 D

$$d_o = 2.7 \times 10^3 \text{ m}$$

$$d_i = 0.025 \text{ m}$$

$$P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = \left(\frac{1}{2.7 \times 10^3 \text{ m}} \right) + \left(\frac{1}{0.025 \text{ m}} \right) = 40 \text{ D}$$

+3

10. Suppose that you would like to find the angle of refraction in a material. You shine a laser through air at an incident angle of 1.8° into the material which has an index of refraction of 1.1. Calculate the angle of refraction in the material.

ANSWER: 1.64°

$$\theta_2 = ?$$

$$\theta_1 = 1.8^\circ$$

$$n_1 = 1$$

$$n_2 = 1.1$$

$$\frac{n_1 \sin \theta_1}{n_2} = \frac{n_2 \sin \theta_2}{n_2}$$

$$\sin \theta_2 = \frac{1 \sin 1.8^\circ}{1.1} = 0.029$$

$$\theta_2 = 1.64^\circ$$

11. A clear-glass light bulb is placed 0.47 m from a thin convex lens that has a focal length of 0.079 m. Calculate the distance from the lens to the location of the produced image.

ANSWER: 0.095m

$$d_o = 0.47\text{m}$$

$$f = 0.079\text{m}$$

$$d_i = ?$$

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\frac{1}{d_i} = \left(\frac{1}{0.079} \right) - \left(\frac{1}{0.47} \right) = \frac{1}{0.095}$$

$$d_i = 0.095\text{m}$$

12. What power of spectacle lens is needed to correct the vision of a far-sighted person whose near point is 0.89 m, so that they may see an object clearly that is 22. cm away? Assume the spectacle lens is held 1.0 cm away from the eye by eyeglass frames.

ANSWER: 3.63D

$$d_o = 0.22\text{m} - 0.01 = 0.21\text{m}$$

$$d_i = 0.89\text{m} - 0.01\text{m} = 0.88\text{m}$$

$$P = \frac{1}{0.21\text{m}} + \frac{1}{-0.88\text{m}}$$

$$P = 3.63\text{D}$$

PHY202 Exam 6 on chapter(s): 25, 26

1. $m = -0.116 \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$
(Chapter 25, Section (6) Image Formation by Lenses - cple_ch25_ex6b)
2. $n_2 = 1.16 \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$
(Chapter 25, Section (3) The Law of Refraction - cple_ch25_ex2)
3. $P = -0.842 \text{ D} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
(Chapter 26, Section (2) Vision Correction - cple_ch26_ex3)
4. $h_i = -1.42 \times 10^{-5} \text{ m} \therefore \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$
(Chapter 26, Section (1) Physics of the Eye - cple_ch26_ex1)
5. $\theta_c = 36.0^\circ \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$
(Chapter 25, Section (4) Total Internal Reflection - cple_ch25_ex4)
6. $m = 1.54 \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$
(Chapter 26, Section (6) Image Formation by Lenses - cple_ch25_ex7)
7. $v = 2.31 \times 10^8 \frac{\text{m}}{\text{s}} \therefore n = \frac{c}{v}$
(Chapter 25, Section (3) The Law of Refraction - cple_ch25_ex1)
8. $P = 17.2 \text{ D} \therefore P = \frac{1}{f}$
(Chapter 25, Section (6) Image Formation by Lenses - cple_ch25_ex5)
9. $P = 40.0 \text{ D} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
(Chapter 26, Section (1) Physics of the Eye - cple_ch26_ex2a)
10. $\theta_2 = 1.64^\circ \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$
(Chapter 25, Section (3) The Law of Refraction - cple_ch25_ex3)
11. $d_i = 0.0950 \text{ m} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$
(Chapter 25, Section (6) Image Formation by Lenses - cple_ch25_ex6a)
12. $P = 3.63 \text{ D} \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
(Chapter 26, Section (2) Vision Correction - cple_ch26_ex4)